Noise Removal versus Signal Loss in Network Filtering of Daily GNSS Time Series

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Abstract

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Keywords: GNSS time series, common-mode error, spatial filtering, transient deformation, signal preservation, deep learning

1. Introduction

Daily position solutions from the Global Navigation Satellite System (GNSS) reach formal uncertainties of about 1 mm in the horizontal components and 3–4 mm in the vertical (Nevada Geodetic Laboratory, 2026). The Nevada Geodetic Laboratory (NGL) processed data from more than 17,000 GPS stations worldwide as of 2018 and distributes the resulting time series openly (Blewitt et al., 2018). Using these series to study crustal deformation, such as slow slip or postseismic afterslip, first requires removing noise, part of which is shared between stations.

Common-mode error is noise shared across many stations of a network. It arises from errors in satellite orbits and clocks, in the realisation of the reference frame and from large-scale environmental loading (Wdowinski et al., 1997; Dong et al., 2006; Kreemer and Blewitt, 2021). Several approaches have been used to estimate and remove it, from stacking (Wdowinski et al., 1997) and principal or independent component analysis (Dong et al., 2006; Gruszczynski et al., 2018; Zhou et al., 2022) to weighted averages and medians of neighbouring stations (Tian and Shen, 2016; Kreemer and Blewitt, 2021). These filters are usually evaluated by how much noise they remove, most often measured as the reduction in the root-mean-square (RMS) scatter

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of the filtered series (Tian and Shen, 2016; Kreemer and Blewitt, 2021; Zhou et al., 2022). Learned filters have been evaluated in different ways. Bachelot et al. (2025) measure noise reduction as the decrease in the average position of the whole network, which should be close to zero in the absence of common-mode noise, and assess signal preservation by comparing 280 slow-slip offsets, each the difference between the mean positions in the ten days after and before an event, measured on raw and denoised series. They note that without ground truth the denoising performance cannot be quantified in an absolute sense. Costantino et al. (2024) add synthetic slow-slip events to noise generated from real GNSS series by principal component analysis and Fourier phase randomisation, which preserves its spatial covariance and spectrum, and report the error of the recovered displacement as a function of the signal-to-noise ratio; on real data, they validate the result by its agreement with tectonic tremor. Mastella et al. (2025), whose method works on single stations, train on synthetic series and evaluate the noise removed from real series against a trajectory-model fit, which they treat as the reference noise, while showing the preservation of a slow-slip transient by example.

The problem with evaluating noise removed using RMS scatter is that the amount of real transient removed is not accounted for. Kreemer and Blewitt (2021) note that the main reason their filter reduces scatter more than traditional stacking is that it absorbs local common-mode transients. How much small signal survives filtering affects the science. In Cascadia, Costantino et al. (2026) find that the moment–duration scaling, b-value and stress-drop distribution of slow-slip events all change with the detection threshold applied to slip inverted from denoised data, a threshold they describe as a proxy for the signal-to-noise ratio. The disagreement over a reported precursory acceleration of displacement before large earthquakes (Bletery and Nocquet, 2023) turned on the same question, whether a common-mode correction removes only noise or also real deformation (Bradley and Hubbard, 2023; Bletery and Nocquet, 2025).

Bachelot et al. (2025) is the closest prior work to this study. They predict each station from its eight nearest stations beyond 400 km, and chose the 400 km empirically, as the shortest distance at which slow-slip offsets in their study area were no longer reduced by denoising. They note that this distance is specific to their region and time window, and it is not clear how either parameter should change for other networks or for transients of other sizes. Distance has been varied before: Tian and Shen (2016) estimated the common mode from stations beyond a series of increasing distance thresholds

and used the change in residual scatter to infer the spatial extent of regional transients. Their thresholds served to locate transients, however, and were judged by scatter, not by how much of a transient the filter removes. In this study we treat both parameters, the exclusion radius and the number of stations averaged, as experimental variables rather than constants: we vary the radius from 0 to 400 km and the number of stations from 16 to the whole network, and measure how each affects the noise removed and the signal kept.

In this study we measure signal loss directly, by planting transients of known size, amplitude and duration in the real residuals of 1,251 stations in and around California. Each filter predicts every station from N other stations, excluding all stations within a radius R of it, and the prediction is subtracted as noise. We run each filter twice, once on the original residuals and once with the transients added. The difference between the two cleaned series is what remains of the planted transient, and its ratio to the transient gives the fraction of signal the filter keeps. The same runs give the noise the filter removes, as the reduction in short-period scatter, and any spurious signal it creates at nearby stations where nothing was planted. Because the transients are planted in real rather than synthetic noise, these measurements have a ground truth without depending on how well a noise model reproduces real data. Testing geodetic methods with known synthetic signals is not new: the SCEC transient-detection exercise compared detection methods on synthetic and real southern California data (Lohman and Murray, 2013). What has not been measured is how the signal a common-mode filter removes depends on the size of the transient relative to the reach of the filter.

We apply this audit to filters from three families: classical filters that include the target station in their own estimate (network stacking, sub-regional principal component analysis and a robust weighted median), filters that exclude all stations within a fixed distance of the target, including the 400 km design of Bachelot et al. (2025), and leave-one-out filters ranging from weighted means and ridge regression to a learned kernel and a space–time attention network, the last also trained with planted transients. Realistic slip events on mapped faults and real earthquake offsets test whether the conclusions drawn from idealised transients hold for real signals. Section 2 describes the data and their processing, Section 3 the audit and its measures, and Section 4 the filters, which are listed in Table 1. Section 5 presents the results, and Section 6 discusses their implications for choosing and evaluating network filters.

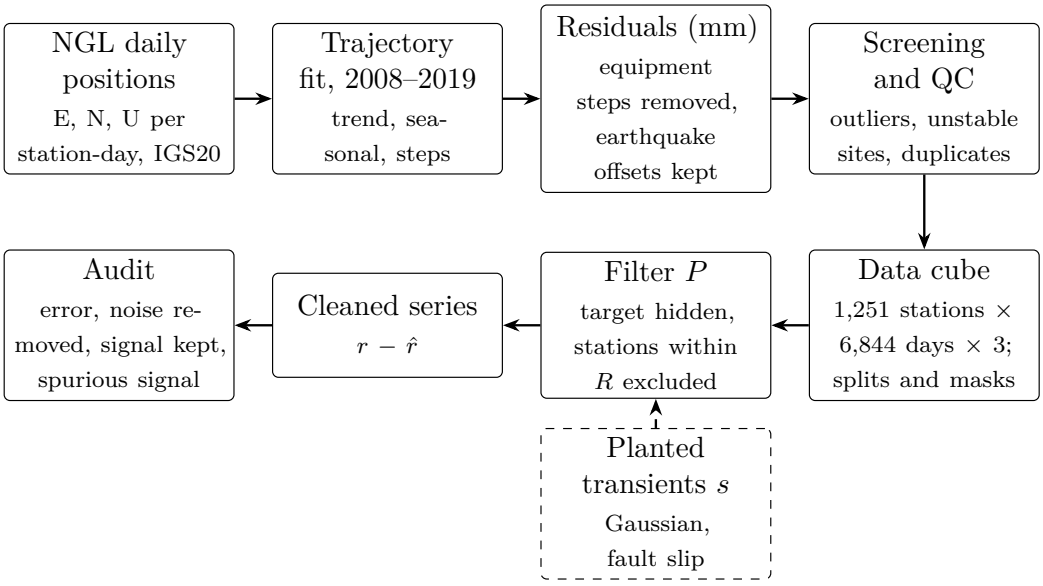


Figure 1: Processing chain from daily GNSS positions to the signal-loss audit.

2. Data and processing

Figure 1 summarises the processing chain. We take daily GNSS positions from NGL and reduce them to residual displacements by removing each station's long-term trajectory. The residuals are screened and arranged in a single array. Each filter then estimates the part of every station's residual that is shared with other stations, either from its N nearest neighbours, from all stations beyond a radius R, or with an established common-mode method, and subtracts it. The cleaned series are scored on the noise removed, the planted transients and real offsets they keep, and the spurious signal they create. The rest of this section describes the data and their preprocessing.

2.1. Data

[Paragraph 2: what one daily solution is.]

[Paragraph 3: station selection and the network (Fig. 2a).]

2.2. From positions to residuals

[Paragraph 4: the trajectory model and why it is fitted on the training years only.]

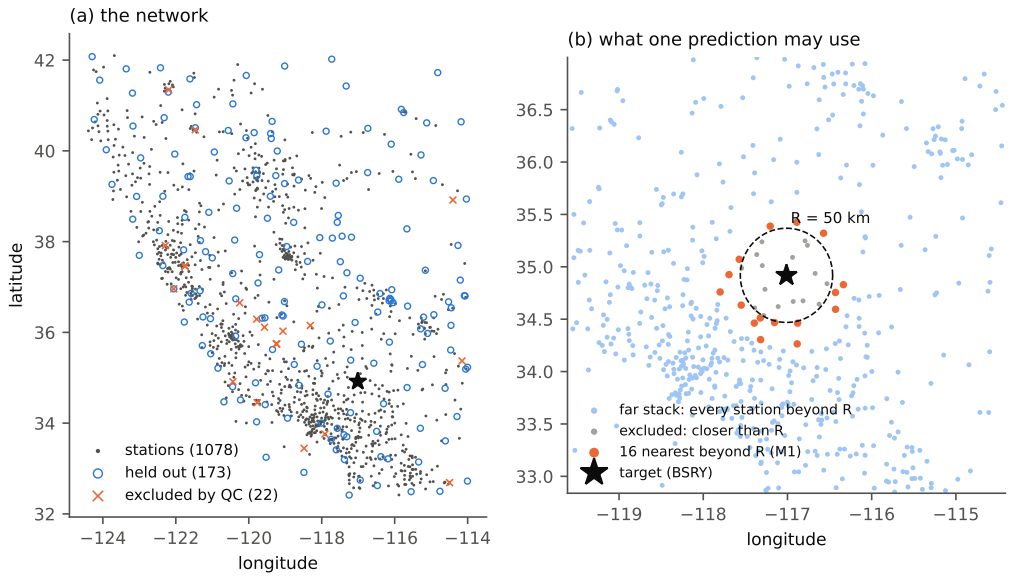


Figure 2: (a) Stations used, held out and excluded. (b) What a prediction for one target may use at $R = 50$ km. [Expand caption.]

$$x_{i,c}(t) = a + v\,(t - t_0) + \sum_{k=1}^{2} \left[s_k \sin(2\pi k t) + c_k \cos(2\pi k t) \right] + \sum_j b_j\, H(t - t_j) + r_{i,c}(t) \tag{1}$$

[Paragraph 5: equipment steps removed, earthquake offsets kept (Fig. 3b).]
[Paragraph 6: outlier screening and quality control.]
[Paragraph 7: the data cube, splits and held-out stations.]
[Paragraph 8: why we filter displacement rather than rates.]

3. The signal-loss audit

3.1. Leave-one-out filtering and the exclusion radius

[Paragraph 1: leave-one-out, the exclusion radius R and leak checks.]

3.2. Four measures

[Paragraph 2: reconstruction error.]
[Paragraph 3: noise removed.]
[Paragraph 4: signal kept.]

$$\rho = \frac{\langle \Delta,\, s \rangle}{\langle s,\, s \rangle}, \qquad \Delta = \Big(r + s - P(r+s)\Big) - \Big(r - P(r)\Big) \tag{2}$$

[Paragraph 5: spurious signal.]

3.3. Planted and real signals

[Paragraph 6: Gaussian transients.]
[Paragraph 7: fault-slip events from the event library.]
[Paragraph 8: real earthquake offsets and Ridgecrest.]

3.4. Protocol

[Paragraph 9: decisions, freeze date, bootstrap, seeds.]

4. Filters

[Paragraph 1: the three families.]
[Paragraph 2: classical and regression filters.]
[Paragraph 3: learned models.]

5. Results

5.1. The trade-off between noise removed and signal kept

[Paragraph 1: the overall trade-off (Fig. 5).]
[Paragraph 2: what the radius does.]

5.2. Number of stations versus distance

[Paragraph 3: count beats distance (Fig. 6).]
[Paragraph 3b: regression on the nearest stations (M3, M4).]

5.3. Realistic transients

[Paragraph 4: near field kept, far field removed (Fig. 7).]
[Paragraph 5: shallow slow slip and the spurious signal it causes.]

5.4. Real earthquake offsets

[Paragraph 6: real offsets confirm the planted results (Fig. 8).]

5.5. Learned models

[Paragraph 7: the learned kernel.]

[Paragraph 8: more neighbours, more noise removed, less signal kept.]

[Paragraph 9: the attention network with and without planted-transient training.]

6. Discussion

[Paragraph 1: what practitioners should do.]

[Paragraph 2: why local filters remove the far field and why it matters.]

[Paragraph 3: relation to Bachelot et al.]

[Paragraph 4: what this means for deep learning.]

[Paragraph 5: limitations and future work.]

7. Conclusions

[Conclusions: write after the discussion.]

Data and code availability

[Data and code availability.]

Declaration of generative AI in the writing process

[AI declaration: write at the end, describing what was actually done.]

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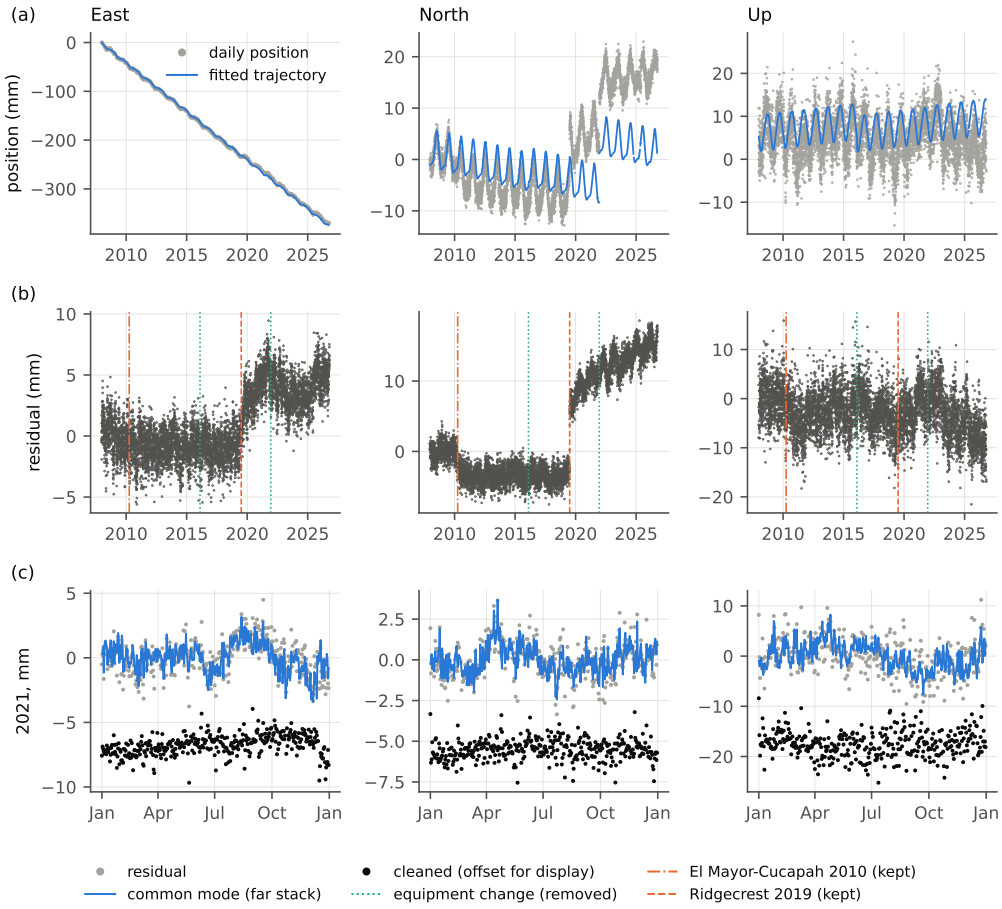


Figure 3: From positions to residuals at station BSRY. [Expand caption: (a) positions and trajectory, (b) residual, (c) one validation year with the network common mode and the cleaned series.]

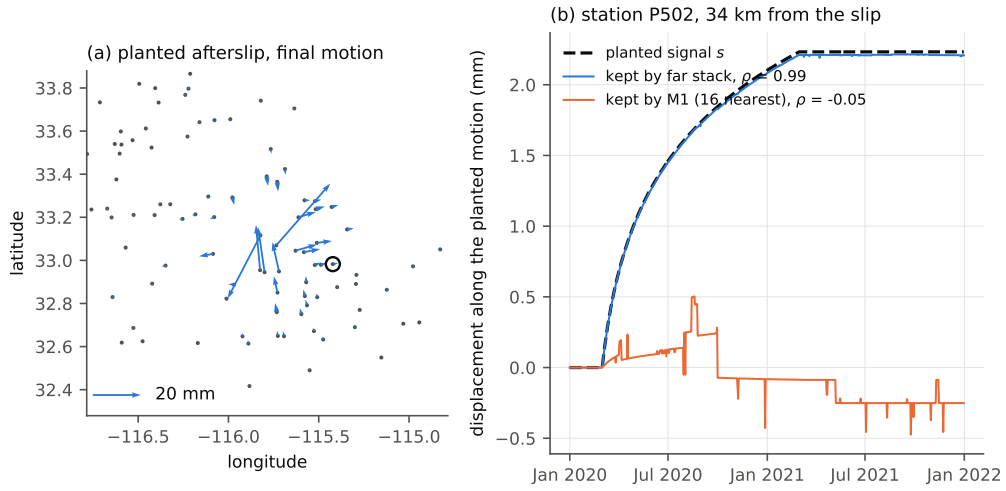


Figure 4: Measuring signal kept. [Expand caption; note that P502 is the median station among 17 at 15–40 km.]

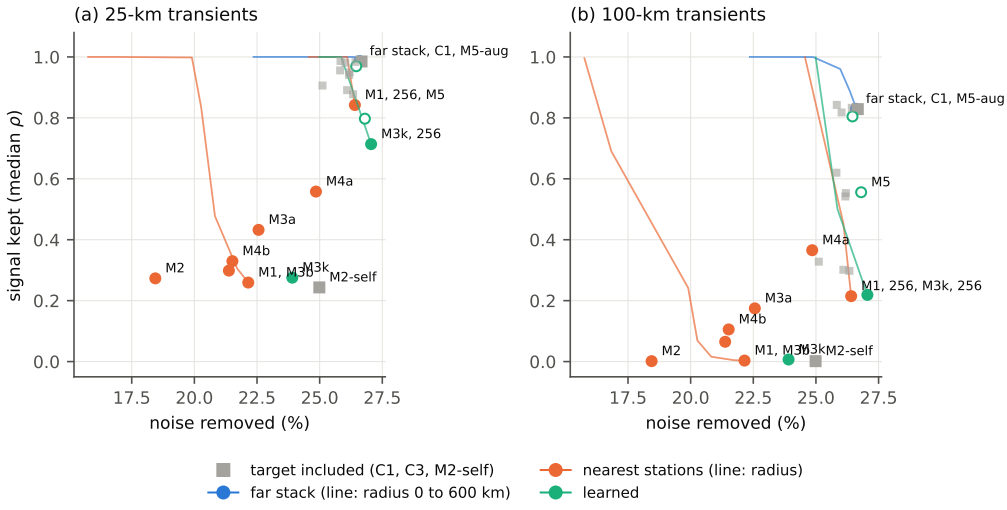


Figure 5: [Caption.]

Table 1: Filters audited. [Check descriptions.]

Name	Description	Parameters
Target-inclusive		
C1	Mean of the whole network, target included	0
C3	Sub-regional probabilistic PCA; 1, 4 or 8 regions × 1–3 components	∼3.7k/comp.
M2-self	Robust weighted median of the 16 nearest and the target	2
Distance-excluded		
Far stack	Mean of every station beyond R; D1 at $R =$ 400 km	0
Leave-one-out over nearest neighbours		
M1	Distance-weighted mean of the 16 (64, 256) nearest beyond R	1
M2	Robust weighted median of the 16 nearest beyond R	2
M3a / M3b	Ridge regression on 16 neighbours, per station / pooled	65/station; 257
M4a / M4b	M3 per time band (under 7, 7–60, 60–400, over 400 days)	4 × M3
Learned		
M3k	Learned kernel: neighbour weight from distance, azimuth and noise level; 16 or 256 neighbours	1,411
M5	Space–time attention: 32 neighbours and the far stack, 64-day window	212,048
M5-aug	M5 trained with planted transients in its inputs	212,048

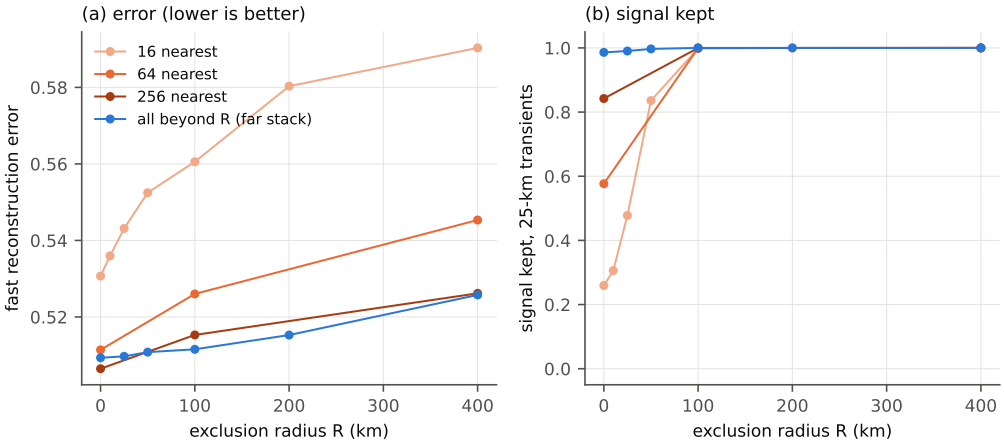


Figure 6: [Caption.]

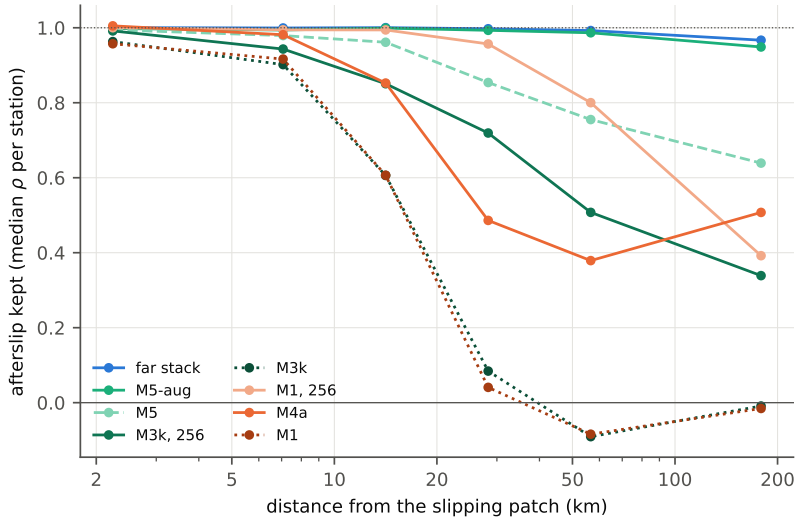


Figure 7: [Caption.]

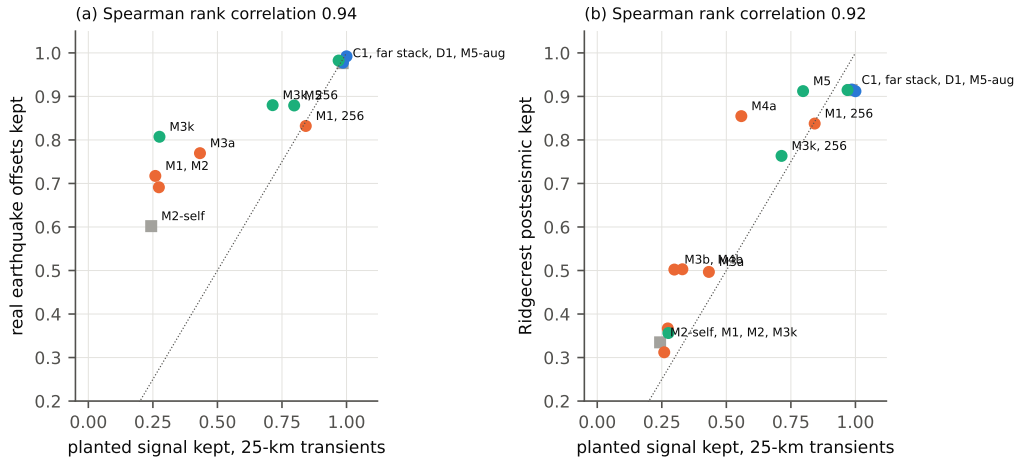


Figure 8: [Caption.]

Table 2: Main results at $R = 0$ unless stated. *Validation years; to be replaced by test-year results after the method freeze.* Error: fast leave-one-out error (lower is better; zero prediction 0.723). Kept: median retention of 25- and 100-km Gaussian transients; afterslip at 15–40 km from the patch; real earthquake offsets.

Filter	Error	Noise removed	Kept 25 / 100 km	Spurious	Afterslip	Real offsets
C1	–	26.7%	0.99 / 0.83	0.11	1.00	0.98
C3 (9 settings)	–	25.1–26.4%	0.88–0.99 / 0.30–0.84	0.27–1.52	0.97–1.00	0.90–0.95
M2-self	–	25.0%	0.24 / 0.00	0.10	0.19	0.60
Far stack	0.509	26.6%	0.99 / 0.83	0.11	1.00	0.98
D1 ($R = 400$ km)	0.526	24.9%	1.00 / 1.00	0.24	1.00	0.99
M1	0.531	22.2%	0.26 / 0.00	0.02	0.09	0.72
M1, 256	0.507	26.4%	0.84 / 0.21	0.10	0.96	0.83
M2	0.558	18.4%	0.27 / 0.00	0.11	0.23	0.69
M3a	0.527	22.6%	0.43 / 0.17	0.04	0.22	0.77
M3b	0.534	21.4%	0.30 / 0.07	0.04	0.24	0.74
M4a	0.517	24.9%	0.56 / 0.37	0.03	0.53	0.75
M4b	0.532	21.5%	0.33 / 0.11	0.04	0.31	0.72
M3k	0.517	23.9%	0.28 / 0.01	0.03	0.11	0.81
M3k, 256	0.499	27.1%	0.71 / 0.22	0.18	0.73	0.88
M5	0.504	26.8%	0.80 / 0.56	0.15	0.87	0.88
M5-aug	0.508	26.5%	0.97 / 0.80	0.17	0.99	0.98